

DAVID BEINHAUER

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RESEARCH INTERESTS

I am computer scientist and bioinformatician interested in the intersection of neuroscience, machine learning and computational biology. I am interested in combination of biological knowledge, computational modeling and experimental data analysis to improve the clinical solutions, such as brain-computer interfaces, sensory restoration or neurodegenerative disease treatment.

EDUCATION

M.Sc. in Bioinformatics, Charles University, Prague, Czech Republic 2022–2025

GPA: **1.33** (scale: 1–4 | 1 is best)

Relevant Coursework: Computational Neuroscience, Spatio-Temporal Modeling of Biological Systems, Cell Biology

B.Sc. in Computer Science, Charles University, Prague, Czech Republic 2019–2022

Specialization: **Artificial Intelligence**

GPA: **1.28** (scale: 1–4 | 1 is best)

Relevant Coursework: Machine Learning, Deep Learning, Statistics, Data Analysis

RESEARCH EXPERIENCE

**Research Assistant, Czech Academy of Sciences,
National Institute of Mental Health**

Dec 2025 – Present

- Working on analysis and comparative evaluation of EEG data from participants receiving psilocybin versus placebo during music listening sessions.
- Investigating inter-subject neural synchrony in response to auditory stimulation.
- Implementing proof-of-concept solution to identify uninformed auditory stimulation patterns in EEG data using IVA decomposition on power wavelet data, using ASSR (auditory steady-state response) dataset as a benchmark. And investigating ASSR response difference between psilocybin and placebo conditions.
- Comparing techniques for identification inter-subject neural synchrony in EEG power wavelet data, using variants of ICA and IVA decomposition, and evaluating their performance on the ASSR dataset.

Modeling Visual Cortex with Biologically Inspired Recurrent Neural Networks

- **Master Thesis:** GitHub (Code) / GitHub (Thesis)
- Developed a biologically realistic model of the primary visual cortex (V1) using modular, multi-layered recurrent neural networks.
- Mapped synthetic neurons on a one-to-one basis with those from a state-of-the-art spiking neural network model of V1.
- Demonstrated that incorporating biological realism is essential for capturing the temporal dynamics of the visual system.

TALKS AND CONFERENCE PRESENTATIONS

- [1] **Beinhauer, D.**, Kraus R, Baroni L, Antolík J (2025). *Bridging Biology and Deep Learning: Modeling Visual Cortex with Structured Recurrent Networks*. Bernstein Conference 2025.
doi: 10.12751/nncn.bc2025.151
- [2] **Beinhauer, D.** (2025, Nov 12) *Bridging Biology and Deep Learning: Modeling Visual Cortex with Structured Recurrent Networks*. Oral research pitch. Young AI Research Forum 2025: AI as Driver for Interdisciplinary Research, Prague, Czech Republic.
- [3] **Beinhauer, D.**, Kraus, R., Baroni, L., Antolík, J. (2025, May 27). *Advancing visual neuroscience with biologically inspired recurrent neural networks*. [Poster presentation by J. Antolík]. The Sixth International Conference on the Mathematics of Neuroscience and AI. Split, Croatia.

INDUSTRY RESEARCH EXPERIENCE

Bioinformatics Internship, MSD Czech Republic

Dec 2025 – Present

- Developing computational solutions within the Bioinformatics team to enhance drug discovery and development processes.
- Customized the loss function for protein structure prediction, improving the in vivo prediction of drug-target interactions.
- Leveraged several scientific computing tools (mosaic, Bindcraft, RFdiffusion, ProteinMPNN, ESM) to analyze and propose alternative mutations in a protein structure.
- Collaborated in the development of the company-based open-source project OVO, which provides tools for protein design. GitHub (OVO)

NLP Internship, MSD Czech Republic

Oct 2023 – Nov 2025

- Developing Natural Language Processing (NLP) applications for the pharmaceutical domain within the Artificial Intelligence team.
- Designed and implemented Large Language Model (LLM)-based solutions for scientific document processing and data retrieval.
- Collaborated with medical writers to integrate LLM techniques into the research and drafting of internal medical documentation.

AWARDS AND HONORS

Travel Grant to the Bernstein Conference

2025

- Awarded a travel grant as an early-career scientist presenting as first author in the poster session at the Bernstein Conference 2025.

Nomination for Dean's Award

2025

- Nominated for one of the best final theses of the academic year at the faculty, an award recognizing outstanding research quality and contribution.

Scholarship for Academic Excellence

2021, 2023

- Awarded twice for ranking among the top 10% of students based on academic performance during both bachelor's and master's studies.

SKILLS

Technical Skills

- Programming Languages: Python, C++
- Machine Learning Frameworks: PyTorch, Scikit-learn, JAX
- Signal Processing & Neuroimaging: MNE-Python, ICA/IVA, wavelet decomposition
- Cloud and Computing Tools: AWS, HPC environments, Docker, Nextflow
- Development Tools: Git, Jira
- Domain Knowledge: Cell Biology, Genomics, Structural Biology, Basic Neurobiology

Languages

- English: Professional proficiency
- Czech: Native speaker

ACADEMIC REFEREES

Mgr. Ján Antolík, Ph.D.

Master Thesis Supervisor

- Email: antolik@ksvi.mff.cuni.cz

Ing. Vlastimil Koudelka, Ph.D.

Research Assistant Supervisor

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Ing. Mgr. Jaroslav Hlinka, Ph.D.

Research Assistant Supervisor

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